

Calculus I Lesson 05

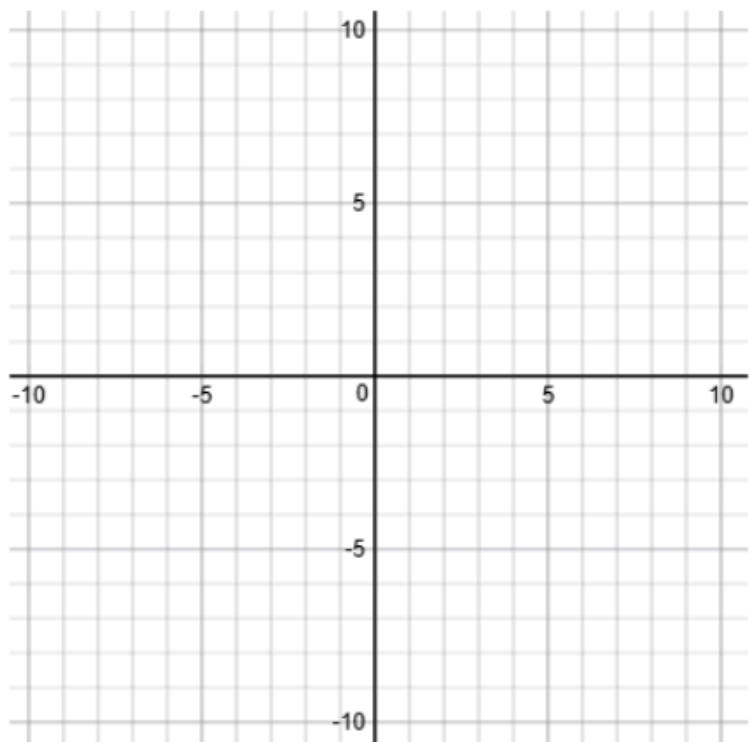
1) Find $\lim_{x \rightarrow 2} \frac{\sqrt{x+2} - 2}{x-2}$.

a) $(\sqrt{x+2} - 2) \cdot (\sqrt{x+2} + 2) = \underline{\hspace{2cm}} ?$

b) $\frac{\sqrt{x+2} - 2}{x-2} \cdot \frac{(\sqrt{x+2} + 2)}{(\sqrt{x+2} + 2)} = \underline{\hspace{2cm}} ?$

c) $\lim_{x \rightarrow 2} \frac{\sqrt{x+2} - 2}{x-2} = \underline{\hspace{2cm}} ?$

d) Graph $f(x) = \frac{\sqrt{x+2} - 2}{x-2}$



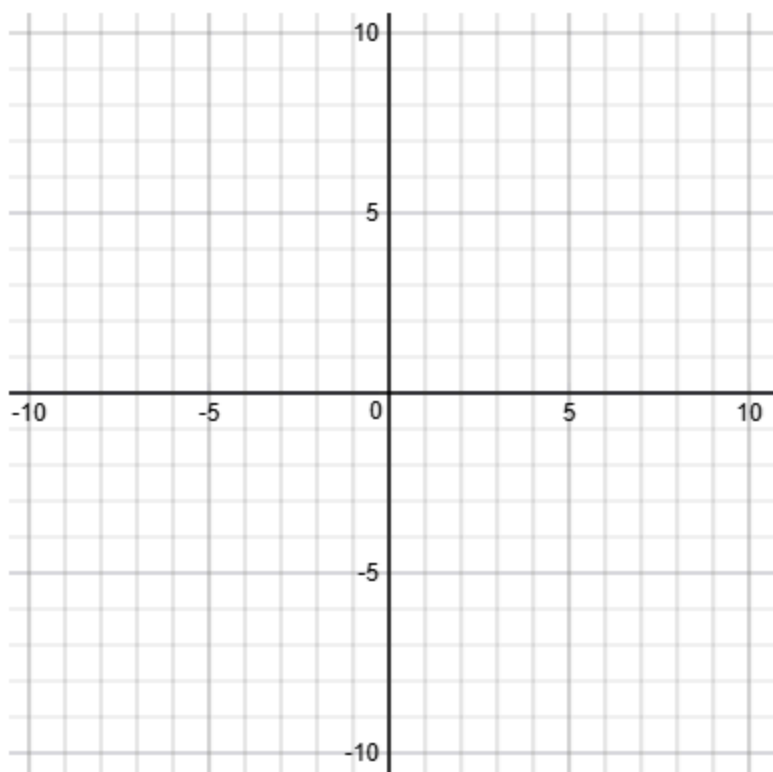
2) Find $\lim_{x \rightarrow 7} \frac{\sqrt{x+2} - 3}{x - 7}$

a) $(\sqrt{x+2} - 3) \cdot (\sqrt{x+2} + 3) = \underline{\quad ? \quad}$

b) $\frac{\sqrt{x+2} - 3}{x - 7} \cdot \frac{(\sqrt{x+2} + 3)}{(\sqrt{x+2} + 3)} = \underline{\quad ? \quad}$

c) $\lim_{x \rightarrow 7} \frac{\sqrt{x+2} - 3}{x - 7} = \underline{\quad ? \quad}$

d) Graph $f(x) = \frac{\sqrt{x+2} - 3}{x - 7}$



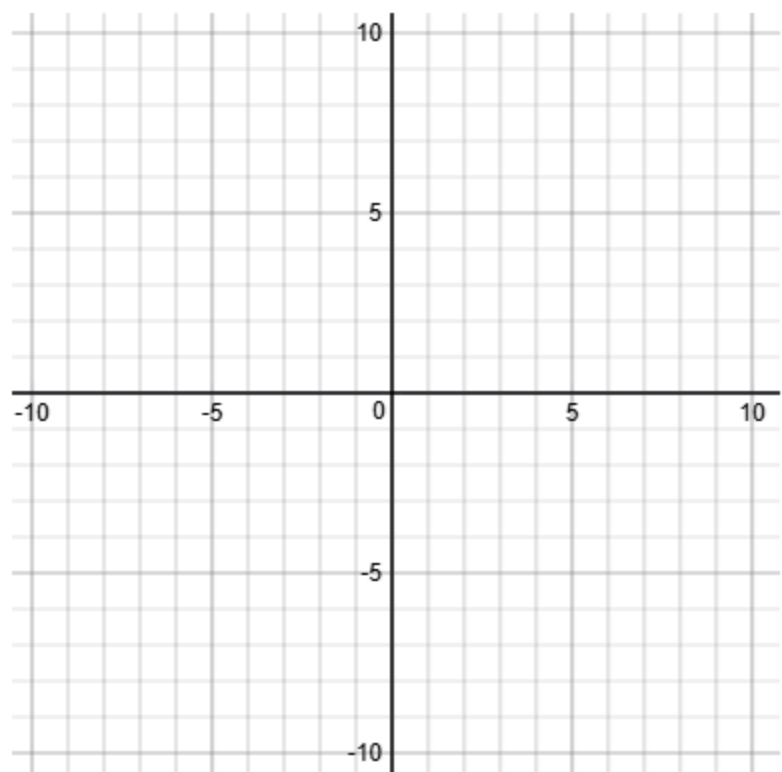
$$3) \lim_{x \rightarrow 6.5} \frac{\sqrt{2x+3}-4}{2x-13}$$

$$a) (\sqrt{2x+3}-4) \cdot (\sqrt{2x+3}+4) = \underline{\quad? \quad}$$

$$b) \frac{\sqrt{2x+3}-4}{2x-13} \cdot \frac{(\sqrt{2x+3}+4)}{(\sqrt{2x+3}+4)} = \underline{\quad? \quad}$$

$$c) \lim_{x \rightarrow 6.5} \frac{\sqrt{2x+3}-4}{2x-13} = \underline{\quad? \quad}$$

$$d) \text{ Graph } f(x) = \frac{\sqrt{2x+3}-4}{2x-13}$$



$$4) \lim_{x \rightarrow 11} \frac{\sqrt{4x+5}-7}{4x-44}$$

$$a) (\sqrt{4x+5}-7) \cdot (\sqrt{4x+5}+7) = \underline{\hspace{2cm}} ?$$

$$b) \frac{\sqrt{4x+5}-7}{4x-44} \cdot \frac{(\sqrt{4x+5}+7)}{(\sqrt{4x+5}+7)} = \underline{\hspace{2cm}} ?$$

$$c) \lim_{x \rightarrow 11} \frac{\sqrt{4x+5}-7}{4x-44} = \underline{\hspace{2cm}} ?$$

$$5) \lim_{x \rightarrow 0} \frac{\sqrt{x+5} - \sqrt{5}}{x}$$

$$a) (\sqrt{x+5} - \sqrt{5}) \cdot (\sqrt{x+5} + \sqrt{5}) = \underline{\quad ? \quad}$$

$$b) \frac{\sqrt{x+5} - \sqrt{5}}{x} \cdot \frac{(\sqrt{x+5} + \sqrt{5})}{(\sqrt{x+5} + \sqrt{5})} = \underline{\quad ? \quad}$$

$$c) \lim_{x \rightarrow 0} \frac{\sqrt{x+5} - \sqrt{5}}{x} = \underline{\quad ? \quad}$$

$$6) \lim_{x \rightarrow 1.5} \frac{\sqrt{2x+3} - \sqrt{6}}{2x-3}$$

$$a) (\sqrt{2x+3} - \sqrt{6}) \cdot (\sqrt{2x+3} + \sqrt{6}) = \underline{\hspace{2cm}} ?$$

$$b) \frac{\sqrt{2x+3} - \sqrt{6}}{2x-3} \cdot \frac{(\sqrt{2x+3} + \sqrt{6})}{(\sqrt{2x+3} + \sqrt{6})} = \underline{\hspace{2cm}} ?$$

$$c) \lim_{x \rightarrow 1.5} \frac{\sqrt{2x+3} - \sqrt{6}}{2x-3} = \underline{\hspace{2cm}} ?$$

